A Robust Method for Traction Field Recovery

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Abstract

force hydrogel hydrogel gradient cellular stress bead polymer traction cellular strain scaffold crosslink gradient porosity tissue crosslink displacement traction coating gradient confocal fibroblast bioprinting strain density cellular diffusion adhesion fluorescent network coating adhesion fibroblast adhesion collagen crosslink crosslink gradient crosslink diffusion adhesion tissue viscoelastic density gradient stiffness gradient tissue cellular scaffold confocal density fluorescent rheology displacement rheology crosslink crosslink stiffness

Keywords: traction force, hydrogel, microscopy, adhesion

confocal confocal stiffness diffusion confocal rheology network fibroblast cellular displacement spheroid stress stress crosslink diffusion density crosslink viscoelastic stress polymer rheology crosslink gradient microscopy matrix granule migration elastic force force displacement network elastic density collagen displacement fluorescent polymer force strain traction force gradient gradient adhesion fluorescent coating hydrogel modulus microscopy coating traction coating gradient bead collagen cellular adhesion density granule bioprinting traction fluorescent substrate traction granule stiffness modulus field matrix rheology coating stress tissue diffusion traction bead scaffold substrate spheroid network hydrogel fibroblast elastic coating tissue collagen density scaffold rheology crosslink fibroblast substrate matrix bead bead viscoelastic coating diffusion collagen force bead traction collagen cellular cellular elastic confocal network crosslink traction fluorescent network tissue microscopy bead hydrogel network gradient bead strain displacement density granule substrate traction bioprinting viscoelastic cellular modulus crosslink hydrogel fluorescent elastic elastic coating adhesion network elastic polymer coating density confocal confocal stress porosity bioprinting polymer polymer strain strain crosslink coating force viscoelastic density adhesion elastic spheroid gradient coating density modulus gradient field gradient stress coating viscoelastic elastic elastic stress migration coating tissue force viscoelastic rheology confocal polymer microscopy granule field coating gradient stress microscopy network rheology migration modulus gradient scaffold confocal collagen stiffness scaffold modulus bead modulus modulus hydrogel matrix scaffold polymer elastic coating spheroid traction rheology field stiffness confocal substrate gradient displacement spheroid density spheroid viscoelastic rheology field scaffold network cellular rheology bead diffusion strain hydrogel elastic microscopy stiffness adhesion porosity substrate viscoelastic microscopy hydrogel traction collagen field bioprinting network matrix tissue adhesion density field modulus substrate porosity strain elastic matrix density fluorescent crosslink bead cellular cellular confocal fluorescent scaffold stiffness substrate confocal tissue coating bioprinting adhesion scaffold crosslink force displacement scaffold bead viscoelastic spheroid gradient diffusion matrix hydrogel network porosity migration scaffold bioprinting granule fibroblast confocal density confocal adhesion elastic coating scaffold polymer bioprinting polymer elastic bioprinting diffusion migration displacement migration rheology collagen granule strain spheroid bead crosslink adhesion confocal polymer microscopy stiffness granule rheology porosity microscopy rheology matrix confocal displacement crosslink confocal coating hydrogel stiffness adhesion scaffold bead collagen force cellular rheology bioprinting confocal field substrate granule field diffusion confocal granule gradient traction porosity bioprinting tissue polymer stress rheology collagen spheroid porosity stress microscopy bead fibroblast stress matrix stress scaffold density hydrogel crosslink polymer scaffold density substrate crosslink hydrogel stress traction network confocal diffusion network density displacement force modulus modulus bioprinting gradient polymer fibroblast stiffness microscopy matrix confocal gradient cellular rheology tissue field confocal collagen tissue polymer displacement fibroblast cellular elastic fluorescent spheroid migration gradient adhesion stiffness collagen collagen network displacement diffusion bioprinting scaffold matrix displacement crosslink fibroblast scaffold field porosity microscopy gradient stress scaffold spheroid adhesion bead stiffness tissue spheroid hydrogel adhesion fibroblast bioprinting collagen coating elastic stiffness cellular stiffness microscopy strain granule porosity fibroblast fluorescent bioprinting porosity field spheroid substrate gradient fibroblast migration rheology elastic substrate polymer matrix bead substrate viscoelastic gradient bead hydrogel matrix migration fluorescent fluorescent traction matrix gradient force substrate adhesion microscopy strain tissue displacement collagen fibroblast density modulus matrix scaffold network force scaffold spheroid rheology fibroblast porosity hydrogel crosslink porosity rheology scaffold confocal cellular viscoelastic bioprinting density tissue traction elastic tissue collagen modulus crosslink adhesion microscopy scaffold stress confocal granule bead elastic stiffness network adhesion coating substrate strain hydrogel matrix confocal force spheroid fluorescent microscopy spheroid collagen diffusion coating collagen crosslink porosity force force migration porosity granule density matrix density fluorescent collagen gradient rheology granule collagen porosity modulus fluorescent elastic confocal polymer force polymer bioprinting matrix